Python Programming

Day	Topics / Activities	Learning Objectives	Assignments
Day 1	- Introduction to Programming - What is Python; Python applications - Installing Python & VS Code - Running Python scripts - print() function - Formatting output - Multi-line strings	- Understand what programming is and how Python is used - Set up a Python development environment - Run a simple Python program - Display formatted text output	- Install Python & VS Code - Write Hello World program - Create a program that prints a short introduction about yourself
Day 2	- Variables: Naming convention - Data types (str, int, float, bool) - Type conversion - input() function - User interaction programs	- Create and use variables to store data - Collect user input - Convert between data types - Build simple interactive programs	- User profile program (name, age, hobby) - Temperature converter program
Day 3	- Arithmetic operators - Comparison operators - Logical operators - Order of operations - Using operators in programs	- Perform mathematical operations in Python - Use comparison operators for logical tests - Combine conditions using logical operators	- Build a basic calculator - Program to compare two numbers and display results
Day 4	- Conditional statements (if, elif, else) - Nested conditions - Boolean logic; Program decision-making	- Write programs that make decisions - Evaluate multiple conditions - Use nested if statements	- Grading system program - Program that determines if a number is odd/even or positive/negative
Day 5	- Mini Project (Week 1 concepts) - Program planning - Writing pseudocode - Debugging basics	- Apply Week 1 concepts to solve real problems - Plan program structure before coding	- Build a simple interactive application such as a quiz game or simple banking system

		- Improve debugging skills	
Day 6	- Loops: for loop and while loop - Iteration; Loop control (break, continue) - Nested loops	- Automate repetitive tasks - Control the flow of loops - Solve problems using loops	- Multiplication table generator - Number guessing game
Day 7	- Functions - Defining functions; - Parameters and arguments - Return values Function reuse - Introduction to *args and **kwargs	- Write reusable code using functions - Break large programs into smaller components	- Build function-based calculator - Convert previous programs into functions
Day 8	- Lists - Creating lists - Indexing and slicing - List methods - Dictionaries - Key-value data storage - Iterating through lists and dictionaries	- Store and manipulate collections of data - Access and modify list and dictionary elements	- Student score manager program; Contact book using dictionaries
Day 9	- Python libraries - Importing modules - Standard libraries - Practical libraries (turtle, qrcode, pyttsx3)	- Understand how libraries extend Python capabilities - Use libraries to build real-world programs	- QR Code generator - Turtle graphics drawing - Text-to-speech program
Day 10	- Weekly Project (Week 2 concepts) - Program design - Combining loops, functions, lists, dictionaries, and libraries	- Integrate programming concepts to solve a practical problem - Improve problem-solving and coding skills	- Build a small application such as a password generator, mini game, or task manager

Day 11	- Introduction to Object-Oriented Programming; Classes and objects - Attributes and methods; The self keyword - Creating simple class-based programs	- Understand the principles of OOP - Create and use classes to model real-world objects	- Create Student class program - Build Car class simulation
Day 12	- GUI Programming with Tkinter (Basics) - Creating windows; Labels; Buttons; Entry widgets; Event handling	- Build simple graphical user interfaces - Handle user interaction through GUI components	- Create login form GUI - GUI program that collects user information
Day 13	- GUI Layouts and Event Handling - Frames - Layout managers (pack, grid, place) - Button events and callbacks	- Design organized graphical layouts; Connect buttons to program logic	- Build GUI calculator
Day 14	- Advanced Tkinter Widgets - Listbox - Checkbutton - Radiobutton - Combobox - Messagebox - Simple menus	- Build interactive applications with multiple widgets	- Create multi-widget GUI application such as a settings panel
Day 15	- Capstone Project - Project testing - Debugging - Project presentation - Code review	- Integrate Python concepts including OOP and GUI - Present a working Python application	- Final project demo - Code submission - Peer feedback